

OC-0616 Local control, metastasis and visual function after brachytherapy in choroidal melanoma.

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Purpose or Objective

Uveal melanoma is the most common primary intraocular tumor. The first-choice treatment in medium-sized tumors is episcleral-plaque brachytherapy.

Our aims are:

To study the efficacy of brachytherapy in choroidal melanoma with the goal of local tumor control and conservation of eyeball and its function.

To study progression-free survival at 5 years from treatment.

Materials and Methods

Retrospective observational study including 100 patients with choroidal melanoma treated with brachytherapy with plaques loaded with Iodine¹²⁵ or Ruthenium¹⁰⁶. Local tumor control and evaluation of metastasis were performed periodically using ocular ultrasound, funduscopy, retinography and body-TC.

Results

Visual function was preserved in 78% of patients. Eyeball was preserved in 81%. Characteristics of patients and tumours are described in Table 1. Local tumor control was achieved in 82%. After 5 years of follow-up, 88% of patients have not developed distant metastasis. 66% of metastasis were localized in liver, 33% in lungs and there was one case with disease affecting lungs and bone at the same time. Enucleation was carried out in 19 patients, of whom 37% of cases were due to treatment toxicity while the remaining 63% of enucleations were due to lack of local control. Metastasis development was higher in enucleated patients than in patients with eyeball conservation ($p=0.047$) (Figure 1). 50% of enucleated patients due to lack of local control developed distant disease in the follow up.

Table 1. General characteristics of patients treated with episcleral-plaque brachytherapy.

VARIABLE	N (%)
N	100
Age, mean (range)	60.07 (23.9 - 86.8)
Sex	
Men	50 (50)
Women	50 (50)
Eye	
Right	53 (53)
Left	47 (47)
Eye colour	
Brown	90 (90)
Blue	4 (4)
Green	6 (6)
Location	
Choroid	90 (90)
Ciliary body / iris	10 (10)
Size (thickness), mean (SD)	5.13 (+/- 2.50)
Plaque type	
BEBIG (¹⁰⁶ Ru)	14 (14)
COMS (¹²⁵ I)	86 (86)
Local control	
Yes	88 (88)
No	12 (12)
Local control assessment method	
Ocular ultrasound	85
Funduscopy	79
Retinography	2
Metastasis	
Yes	18 (18)
No	82 (82)
Enucleation	
Yes	19 (19)
No	81 (81)
Reason for enucleation	
Local recurrence	11 (57.9)
Growth	1 (5.3)
Retinal detachment	3 (15.8)
Neovascular glaucoma	1 (5.3)
Vitreous Haemorrhage	2 (10.5)
Local complications	1 (5.3)
Visual function	
No	22 (22)
Yes	40 (40)
SL	5 (5)
HM	8 (8)
CF	25 (25)

SD: standard deviation. SL: sensing light. HM: hands movement. CF: count fingers.